More women owners, or more women working at home on contract? Women's enterprise ownership in India, 2015–16 to 2023–24

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Abstract. The share of India's unincorporated non-agricultural enterprises owned by women rose sharply in the last decade, a change often read as a gain in women's entrepreneurship. Using the unit-level data of the NSS 73rd round (2015–16) and the Annual Survey of Unincorporated Sector Enterprises 2023–24, we ask where the new women-owned enterprises are. We decompose the change in the number of female-proprietor enterprises into cells defined by activity, location (inside or outside the owner's home), dependence on a contractor and rural or urban area, and split the change in the female share into a within-cell and a between-cell part. We then compare the returns of these enterprises, measured as gross value added per worker, with the earnings women could expect outside them, estimated from the Periodic Labour Force Survey 2023–24 (a woman casual worker earned Rs [pending PLFS run] a day). The female-proprietor share in the harmonised coverage rose from [pending ASUSE/NSS73 data] to [pending ASUSE/NSS73 data] per cent; cells of home-based enterprises account for [pending ASUSE/NSS73 data] per cent of the rise in the number of women owners and contract cells for [pending ASUSE/NSS73 data] per cent. The median female-proprietor enterprise generates [pending ASUSE/NSS73 data] rupees of value added per worker per month, and [pending ASUSE/NSS73 data] per cent generate less than 25

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days of the casual wage. Counts of women owners therefore overstate the growth of women's entrepreneurship in the sense of independent, remunerative enterprise; survey statistics on ownership should be read alongside location, contract dependence and returns.

Keywords: women's entrepreneurship; home-based work; putting-out; informal enterprises; India

JEL codes: J16, L26, O17, J22

Plain English Summary. More of India's small non-farm businesses are now owned by women. We use two large government surveys, from 2015–16 and 2023–24, to see what kind of businesses these are. We find that much of the growth is in very small businesses run from home, often making goods for a contractor who supplies the work and buys the output, and that many of them earn less per worker than a woman could earn as a daily wage labourer. The number of women business owners is a poor guide to women's economic independence. Policy that counts women owners should also track where they work, who they work for, and what they earn.

1 Introduction

India's official enterprise surveys show a fast rise in women's ownership of unincorporated non-agricultural enterprises. In the NSS 73rd round (2015–16), about one in five enterprises in our harmonised coverage were owned by a woman proprietor; in the Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24 the figure was [pending ASUSE/NSS73 data] per cent (National Sample Survey Office, 2016; National Statistics Office, 2025). The published factsheet for 2023–24 reports a rise from 22.9 to 26.2 per cent in a single year. Because these surveys are the main source on India's informal enterprises, the rise is widely cited as evidence of growing women's entrepreneurship.

This paper asks a simple question: where are the new women-owned enterprises, and what do they earn? Ownership in these surveys is a statement about who runs the enterprise; it says nothing about whether the enterprise sells on its own account, is run from a separate workplace, or earns more than the owner's alternatives. A woman who stitches garments

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at home for a contractor who supplies the cloth and buys every piece is recorded as the proprietor of an own-account manufacturing enterprise. So is a woman who runs a shop. The two are different economic situations, and the literature on home-based and subcontracted work has long argued that the first is closer to dependent wage work than to entrepreneurship (Kabeer, Mahmud, and Tasneem, 2018; Kesar, 2024).

We make three contributions. First, we bring the two surveys onto a common coverage and decompose the change in the number of female-proprietor enterprises into cells defined by broad activity, location of the enterprise (in the owner's home or outside), dependence on a contractor (work done solely or mainly on contract) and rural or urban area. A shift-share split separates growth of women's ownership within cells from reallocation of enterprises across cells. Second, we profile the growth margin, the enterprises that account for most of the rise. Third, we benchmark the returns of these enterprises against the outside option of the women who run them, the earnings of women casual workers and self-employed women in the Periodic Labour Force Survey (PLFS) 2023–24 (National Statistics Office, 2024). This benchmark turns a statement about ownership into a statement about welfare: an enterprise that generates less value added per worker than the casual wage is a poor route out of low-paid work, whatever its legal form.

The closest studies are Bose (2023), who documents a large productivity penalty for women-owned home-based manufacturing firms in the 2015–16 data, Kesar (2024), who shows that women-owned informal enterprises are far more likely to work under subcontract, and Mohanan and Kar (2025), who link the recent growth of unorganised manufacturing to women's job work. Shuvam and Mohanty (2023) classify proprietorship by necessity and opportunity motives across earlier rounds. None of these studies follows the change from 2015–16 to the post-pandemic ASUSE rounds across manufacturing, trade and services, decomposes it by home location and contract dependence together, or compares enterprise returns with women's outside option. The paper also speaks to work on gender gaps in microenterprise returns (Fafchamps et al., 2014; Mel, McKenzie, and Woodruff, 2009), on credit and ownership in India (Chaudhuri, Sasidharan, and Raj, 2020), on the local determinants of female entrepreneurship (Ghani, Kerr, and O'Connell, 2013) and on the stagnation and measurement

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of women's work in India (Deshpande and Kabeer, 2021; Klasen and Pieters, 2015).

2 Data

2.1 Enterprise surveys

The NSS 73rd round (July 2015 to June 2016, Schedule 2.34) and ASUSE 2023–24 (October 2023 to September 2024) are nationally representative surveys of unincorporated non-agricultural enterprises, excluding construction, run by the National Statistics Office with a two-stage stratified design (National Sample Survey Office, 2016; National Statistics Office, 2025). ASUSE succeeded the earlier rounds with a near-identical schedule, so the items we use are asked in nearly the same form in both (we map the codes item by item; see the code repository): type of ownership (with separate codes for male, female and transgender proprietors), location of the enterprise (within household premises or outside), whether the enterprise is an own-account enterprise (no regular hired worker) or an establishment, whether it works on contract and, if so, whether solely or mainly for a contractor or mainly for its own customers, whether the contractor supplies raw material or equipment, months of operation, usual hours a day, and the number of working owners, hired workers and unpaid helpers by sex. The NSS 73rd round surveyed 2,90,113 enterprises and ASUSE 2023–24 surveyed 4,98,024 establishments.

We use the final multipliers supplied with each data set, the average of the two semi-round multipliers for the 2015–16 round as prescribed in its estimation procedure, and the stratum and first-stage unit identifiers for the design-based standard errors. To compare like with like we restrict both rounds to manufacturing (NIC 2008 divisions 10–33), trade (45–47) and other services (49–96 except 84), and drop residential renting (NIC 68108), which ASUSE tabulates separately. "Female-proprietor enterprise" means proprietary ownership by a woman; partnerships, self-help groups, cooperatives and other forms are kept as a third category.

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2.2 Returns and the outside option

Gross value added (GVA) is the survey's own item, computed from receipts and operating expenses for the reference period and annualised following the published procedure (by twelve for perennial enterprises and by the reported months of operation for seasonal and casual ones). We work with monthly GVA per worker, where workers include working owners and unpaid helpers, and restrict the returns analysis to market enterprises. GVA per worker is an upper bound on the owner's earning per worker, because it is before rent, interest and the cost of capital.

The outside option comes from the PLFS 2023–24 unit data. We take the average daily earnings of women in casual wage work in current weekly status (Rs [pending PLFS run] in rural areas and Rs [pending PLFS run] in urban areas; Rs [pending PLFS run] overall, against Rs [pending PLFS run] for men), and the average gross earnings over the last 30 days of self-employed women (Rs [pending PLFS run], against Rs [pending PLFS run] for men). A month of casual work at 25 days is therefore worth Rs [pending PLFS run]; we report 22 and 30 days as sensitivity bounds.

2.3 Validation

Before any analysis, each estimate that has a published counterpart is reproduced from the unit data. The PLFS benchmark reproduces the published quarterly all-India average wage of women casual workers in current weekly status within Rs [pending PLFS run] a day in rural areas and within Rs [pending PLFS run] in urban areas in every quarter (Rs [pending PLFS run] for India as a whole), the male all-India wage within Rs [pending PLFS run], and the published 30-day earnings of self-employed women within [pending PLFS run] per cent (Appendix Table A1). For the enterprise surveys we reproduce the number of enterprises, the female-proprietor share, GVA per worker and GVA per enterprise for each round (Table A2); the largest gap is [pending ASUSE/NSS73 data].

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3 Method

Let c index cells formed by activity (manufacturing, trade, other services), location (home, outside), contract dependence (solely or mainly on contract, otherwise) and area (rural, urban), N_{ct} the weighted number of enterprises in cell c in year t, F_{ct} the number with a female proprietor, $e_{ct} = N_{ct}/\sum_c N_{ct}$ the cell share and $s_{ct} = F_{ct}/N_{ct}$ the female share.

The *count decomposition* reports $\Delta F_c/\sum_c \Delta F_c$, the share of the rise in women owners that each cell accounts for. The *shift-share decomposition* of the change in the overall female share $S_t = \sum_c e_{ct} s_{ct}$ is

$$\Delta S = \sum_c \bar{e}_c \, \Delta s_c + \sum_c \bar{s}_c \, \Delta e_c,$$

with bars denoting the average of the two years; the first term is the within-cell change and the second the change due to growth of cells where women's ownership is high. If the rise were driven by women entering independent enterprise across the board, the within term would dominate in outside, own-market cells; if it were driven by the growth of home-based contract work, the between term and the home and contract cells would dominate.

For returns, we report the weighted median and quartiles of monthly GVA per worker by ownership type, home location and contract dependence, the share of enterprises whose GVA per worker falls below 22, 25 and 30 days of the woman casual wage, and GVA per worker-hour. We then estimate, with survey weights and design-based standard errors, regressions of log GVA per worker on a female-proprietor indicator, urban location, NIC two-digit and state fixed effects (model 1), adding home location, contract dependence, own-account status, log hours a day and log months a year (model 2), and interactions of the female indicator with home and contract (model 3). The change in the female coefficient from model 1 to model 2 measures how much of the raw gap is accounted for by where and how women's enterprises operate, in the spirit of Oaxaca (1973). These regressions describe; they do not identify the effect of ownership on returns.

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4 Results

4.1 Where the rise came from

Table 1 (cells accounting for the change in the number of female-proprietor enterprises, 2015–16 to 2023–24) will be generated from p11_results.rds once the unit data are available.

The number of female-proprietor enterprises in the harmonised coverage rose from [pending ASUSE/NSS73 data] lakh to [pending ASUSE/NSS73 data] lakh. The largest single cell is [pending ASUSE/NSS73 data], which accounts for [pending ASUSE/NSS73 data] per cent of the rise. Taken together, home-based cells account for [pending ASUSE/NSS73 data] per cent, and contract cells for [pending ASUSE/NSS73 data] per cent. In the shift-share split, the female share rose by [pending ASUSE/NSS73 data] percentage points, of which [pending ASUSE/NSS73 data] points is within cells and [pending ASUSE/NSS73 data] points is due to the changing mix of cells.

4.2 The growth margin

Table 2 (profile of female- and male-proprietor enterprises by year) will be generated from p11_results.rds.

In 2023–24, [pending ASUSE/NSS73 data] per cent of female-proprietor enterprises operated from the owner's home, against [pending ASUSE/NSS73 data] per cent of male-proprietor enterprises, and [pending ASUSE/NSS73 data] per cent worked solely or mainly on contract, against [pending ASUSE/NSS73 data] per cent. Among female own-account enterprises at home, the share mainly on contract changed from [pending ASUSE/NSS73 data] to [pending ASUSE/NSS73 data] per cent, and the share in manufacturing from [pending ASUSE/NSS73 data] to [pending ASUSE/NSS73 data] per cent.

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4.3 Returns against the outside option

Table 3 (monthly GVA per worker by ownership, location and contract dependence, 2023–24, against the PLFS benchmark) will be generated from p11_results.rds.

The median female-proprietor enterprise generates Rs [pending ASUSE/NSS73 data] of value added per worker per month, against Rs [pending ASUSE/NSS73 data] for male proprietors and Rs [pending PLFS run] for 25 days of women's casual work. [pending ASUSE/NSS73 data] per cent of female-proprietor enterprises fall below that benchmark, against [pending ASUSE/NSS73 data] per cent of male-proprietor enterprises.

In the regressions (Table 4), the raw conditional gap for female proprietors is [pending ASUSE/NSS73 data] per cent. Controlling for home location, contract dependence, own-account status, hours and months, it is [pending ASUSE/NSS73 data] per cent. Working at home is associated with [pending ASUSE/NSS73 data] per cent lower GVA per worker and contract dependence with [pending ASUSE/NSS73 data] per cent.

Table 4 (survey-weighted regressions of log monthly GVA per worker) will be generated from p11_results.rds.

5 Discussion

This section will be written once the results above are generated from the unit data. Its content depends on the sign and size of the within and between terms and on the returns benchmark, and no conclusion is drawn in advance.

Three caveats apply. First, eight years separate the rounds, and the period includes the 2016 demonetisation, the introduction of the goods and services tax in 2017 and the pandemic; the comparison describes the net change and does not attribute it to any of them. The published one-year rise in the female-proprietor share between ASUSE 2022–23 and 2023–24 (22.9 to 26.2 per cent) also suggests that part of the change is recent. Second, GVA per worker is not the owner's income: it includes the return to capital and is shared with unpaid helpers. Third, the PLFS casual wage is the earning of women who found casual work, and

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may overstate what the average woman proprietor could earn outside her enterprise.

6 Conclusion

To be written once the results are generated.

Statements and Declarations

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Competing interests. The author has no competing interests to declare.

Data availability. The unit-level data of the NSS 73rd round, ASUSE 2023–24 and PLFS 2023–24 are available from the Ministry of Statistics and Programme Implementation at microdata.gov.in on registration. The code that produces every number in the paper is available from the author and will be deposited in a public repository on acceptance.

Use of AI tools. Claude (Anthropic) and other AI tools assisted with writing and checking code, checking references and drafting text. The author verified all code, results and references and takes full responsibility for the content.

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Appendix

Table A1 (validation of the PLFS benchmark) will be generated from plfs_benchmark.rds.

Table A2 (validation of the enterprise surveys against published estimates) will be generated from p11_results.rds.

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